XGBoost Regressor

Dataset

House	Size	Age	price
H_1	1000	20	50
H2	1200	15	55
H3	1500	10	70
H4	1800	5	85

Step 1: Initial prediction

mean target $= (50+55+70+85)/4 = 65$

Step 2: Calculate gradient

$$g_i = \hat{y} - y_i$$

House	Actual (y_i)	prediction ($\hat{y}$)	Gradient
H_1	50	65	+15
H_2	55	65	+10
H_3	70	65	−5
H_4	85	65	−20

Step 3: Hessians

For squared error

$h_i = 1$ for every row

Step 4: Find first split

try size < 1400

Left Node

$H_1, H_2 = 15, 10$

Sum $(G_L) = 25$

$H_L = (1+1) = 2$

Right Node

$H_3, H_4 = -5, -20$

$G_R = -25$

$H_R = 2$

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Step 5: Calculate Gain $= \frac{1}{2}\left(\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_L+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right) - \gamma$

$$\text{Gain} = \frac{1}{2}\left(\frac{(25)^2}{2+1} + \frac{(-25)^2}{2+1} - \frac{(0)^2}{5}\right) - \gamma$$

assuming $\lambda = 1$
$\gamma = 0$

208.3

Good split

Step 6 Split left node again

House	Gradient
H_1	15
H_2	10

Try split
Age <18

left-left node
$H_2 (G_L) = 10$
Hessian = 1

left-Right node
$H_1 (G_R) = 15$
Hessian = 1

$$\text{Gain} = \frac{1}{2}\left(\frac{(10)^2}{1+1} + \frac{(15)^2}{1+1} - \frac{(10+15)^2}{1+1+1}\right) - \gamma$$

$$\frac{1}{2}(50 + 112.5 - 208.3) - 0$$

-22.91

Bad Split

Try split left Node
Age < 8

Right left
$H_4 = G = -20$

Righ - Righ
$H_3 = G = -5$

$$Gain = \frac{1}{2}\left(\frac{(-20)^2}{1+1} + \frac{(-5)^2}{1+1} - \frac{(-20-5)^2}{1+1+1}\right) - \gamma$$

$$\frac{1}{2}(200 + 12.5 - 208.3) - 0$$

$$2.08$$

+ve split

Final tree

Size 1400
Leaf A   Age<8
Leaf B   Leaf C

Step 8 - Compute leaf value

leaf A = $G = 25$
$H_2 = 2$

$$\omega = \frac{-G}{H+\lambda} = \frac{-25}{2+1} = -8.33$$

$$\text{leaf } B = \frac{-(-20)}{1+1} = 10$$

$$\text{leaf } C = \frac{-(-5)}{1+1} = 25$$

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YOUVA

Update predition

$H_2, H_1 = \quad F_0 + \eta F_1(x)$ $\qquad$ $H_4 = 64 + 0.1(10)$

$\eta = 0.1$ $\qquad$ $= 66$

$65 + 0.1(-8.33)$

64.17

$H_3 = 65 + 0.1(2.5)$

65.25

House	Prediction
H_1	64.17
H_2	64.17
H_3	65.25
H_4	66

find gradient repeat the process